

Solve decimal problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether? Copy or complete the first step.

2. Calculate $34.75 + 8.906$.

3. Calculate $50 - 17.485$.

4. Miro says: Miro lines up the final digits instead of the decimal points. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether? Copy or complete the first step.
Answer 8.975 km.
2. Calculate $34.75 + 8.906$.
Answer 43.656.
3. Calculate $50 - 17.485$.
Answer 32.515.
4. Miro says: Miro lines up the final digits instead of the decimal points. Is this correct? Explain.
Answer Align ones with ones and decimal points vertically so each place-value column stays correct.

Solve decimal problems

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?

6. Check this result using an inverse, estimate or known fact: Calculate $50 - 17.485$.

2. Calculate $34.75 + 8.906$.

7. Prove or explain your reasoning: Calculate 6.408×7 .

3. Calculate $50 - 17.485$.

8. Identify and correct this misconception: Miro lines up the final digits instead of the decimal points.

4. Solve this independently and show every stage: A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Calculate $34.75 + 8.906$.

10. Write and solve a similar question to: Calculate $50 - 17.485$.

Teacher answers and checking prompts

1. A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?
Answer 8.975 km.
2. Calculate $34.75 + 8.906$.
Answer 43.656.
3. Calculate $50 - 17.485$.
Answer 32.515.
4. Solve this independently and show every stage: A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?
Answer 8.975 km.
5. Use a second method or representation for: Calculate $34.75 + 8.906$.
Answer Expected result: 43.656. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Calculate $50 - 17.485$.
Answer Expected result: 32.515. A valid check must be shown.
7. Prove or explain your reasoning: Calculate 6.408×7 .
Answer 44.856. The explanation must show why the method works.
8. Identify and correct this misconception: Miro lines up the final digits instead of the decimal points.
Answer Align ones with ones and decimal points vertically so each place-value column stays correct.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Calculate $50 - 17.485$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve decimal problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Calculate 6.408×7 .

6. Create a new example that tests the same idea as: Calculate $50 - 17.485$.

2. Prove the answer to this question, rather than only calculating it: A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Calculate $34.75 + 8.906$.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 32.515.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro lines up the final digits instead of the decimal points.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Calculate 6.408×7 .
Answer 44.856. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: A runner completes 3 laps of 2.375 km and then 1.85 km. How far altogether?
Answer Expected result: 8.975 km. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Calculate $34.75 + 8.906$.
Answer Expected result: 43.656. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 32.515.
Answer One valid reconstruction is: Calculate $50 - 17.485$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro lines up the final digits instead of the decimal points.
Answer Align ones with ones and decimal points vertically so each place-value column stays correct.
6. Create a new example that tests the same idea as: Calculate $50 - 17.485$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.